

Marie Roth

EDUCATION

Apr. 2022 – Mar. 2025

PhD in Mathematics

RPTU Kaiserslautern–Landau (Germany) and Aix-Marseille University (France)

- Thesis title: Character sheaves and modular representations of finite reductive groups
- Supervisors: Olivier Dudas and Gunter Malle
- Final grade: Outstanding, magna cum laude
- Dissertation award of the "Freundeskreis der RPTU in Kaiserslautern"

2020 – Feb. 2022

Master's degree in Mathematics

Ecole Polytechnique Fédérale de Lausanne (Switzerland)

- Thesis title: Ennola duality in subgroups of classical groups supervised by David Craven and Donna Testerman
- Final grade: 5.83/6 (6/6 Master's thesis)

2017 – 2020

Bachelor's degree in Mathematics

Ecole Polytechnique Fédérale de Lausanne (Switzerland)

- Third year in exchange at the University of Waterloo, Canada.
- Final grade: 5.31/6 (6/6 Bachelor's thesis)

2014 – 2017

Baccalauréat Scientifique

Lycée Présentation de Marie (France)

- Final grade: 19.75/20 "mention très bien" and magna cum laude

2011 – 2017

Euler class graduate

Ecole Polytechnique Fédérale de Lausanne (Switzerland)

- Intended for children with a very high mathematical intuition. Cover the whole maths program until the end of the first year at EPFL.

EMPLOYMENT

Jun. 2025 – now

Senior Research Assistant

University of East Anglia (United Kingdom) with Vanessa Miemietz

Apr. 2022 – Jun.2025

Research Assistant

RPTU Kaiserslautern–Landau (Germany) with Gunter Malle

- funded by SFB–TRR 195 "Symbolic Tools in Mathematics and their Application" of the German Research Foundation (DFG)

2018 – 2020

Teaching assistant

Ecole Polytechnique Fédérale de Lausanne (Switzerland)

2014 – 2019

Tutor

Ecole Polytechnique Fédérale de Lausanne (Switzerland)

Last updated: 6th September 2026

TEACHING

Now — **Bachelor's supervisor**
Supervising Bachelor's project on "Lie Algebras"

Mar. 2026-Mai 2026 — **Co-lecturer**
Co-taught "Galois Theory" with Vanessa Miemietz

2022-2024 — **Teaching assistant**
Taught and planned various exercise classes (Character theory, Represensation Theory)

2020 – 2021 — **Teacher**
Lycée Presentation de Marie (France)
- Prepared final year students for university mathematics.
- Taught online and on-site, prepared lessons and exercises.
- Motivated the students to learn and think in a different way, by giving them exercises to do together, quizzes during the class (Kahoots), activities to do maths in other contexts.

2018 – 2020 — **Teaching assistant**
Ecole Polytechnique Fédérale de Lausanne (Switzerland)
- Linear Algebra, Physics, Consolidation (Mise à Niveau), Information Computation and Communication, Lie Algebras, Mathematical Logic.
- Answered questions, marked exams and corrected exercises.

2014 – 2019 — **Tutor**
Ecole Polytechnique Fédérale de Lausanne (Switzerland)
- Tutored for Cours Euler (exercise classes).

Jul. 2017 — **Euler Camp Counsellor**
Ecole Polytechnique Fédérale de Lausanne (EPFL)
- Supervised children whilst they were doing various activities: mathematics, hiking and games.

ORGANISATIONAL EXPERIENCE

Now — Organising the ANTLR (Algebra, Number Theory, Logic and Represensations) seminar at the University of East Anglia

Mai 2026w — Co-organised the "International Women in Mathematics Day" at the University of East Anglia

Jan. 2026 — Co-organised the conference "Categorification" at the University of East Anglia

Sept. 2025-Jan. 2026 — Organised the reading group on character sheaves

2022–Jun. 2025 — Member of the gender committee of the SFB–TRR 195

2023 — Co-organised the workshop "How to advance your research career and get funding" in Kaiserslautern (Germany)

2021 — Organised a reading group with other students to learn about the representation theory of semisimple Lie algebras, with the help of Donna Testerman.

PUBLICATIONS AND PREPRINTS

Publications

M. Roth. "Unitriangularity of decomposition matrices of the unipotent ℓ-blocks for simple adjoint exceptional groups". Ann. Represent. Theory 2.4 (2025), pp. 537–574

Last updated: 6th September 2026

Preprints

V. Miemietz, M. Roth and D. Tubbenhauer. The subregular and submaximal p-cells. 2026. arXiv: `2608.19798 [math.RT]`

SELECTED TALKS

Invited talks

Mar. 2026 Unitriangularity of decomposition matrices: the simple adjoint finite groups of exceptional type, 'Algebra seminar, University of Birmingham" (United Kingdom)

Nov. 2025 Character sheaves and their restriction to mixed conjugacy classes, "Oxford algebra seminar" (United Kingdom)

Nov. 2024 On the unitriangularity of decomposition matrices of finite reductive groups, "Groups and Representation Theory in Lausanne 2024" (Switzerland)

Contributed talks and posters

Dec. 2024 Poster: Are decomposition matrices of finite groups of Lie type unitriangular? "Women in Algebra and Symbolic Computation III" in Bad Dürkheim (Germany)

Jul. 2024 On the unitriangularity of decomposition matrices of finite reductive groups, "Group Theory in Florence IV" (Italy)

Mar. 2024 Why are exceptions so valuable: a quick introduction to the Moonshine conjecture, "Séminaire doctorant CPT/I2M à Luminy" (France)

SELECTED CONFERENCES

Aug. 2025 "AGGITatE 2025: Groups, Representations and Cohomology" in Essex (United Kingdom)

Jul. 2025 "Modular Lie theory" in York (United Kingdom)

Mar. 2024 "Colloque tournant du Réseau Thématique «Algèbre»" in Marseille (France)

Aug. 2023 "LMS-Bath Summer School and Symposium on Geometric and Categorical Representation theory" in Bath (United Kingdom)

Jun. 2023 "Representations of Reductive Groups" in Aachen (Germany)

Sep. 2022 "Young Group Theorists Workshop" in Les Diablerets (Switzerland)

May 2022 Workshop "Counting conjectures and beyond" in Cambridge (United Kingdom)

ADDITIONAL SKILLS

Languages

French (Mother tongue), English (Fluent PCE C2), German (Academic level B1), Spanish (beginner A2)

Programming

LaTeX (advanced), GAP (intermediate), Python (beginner), C++ (beginner), Matlab (beginner)

REFERENCES

Available on request